Class 19: Pertussis Mini Project

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Background

Pertussis is a bacterial lung infection also known as Whooping cough. Let's begin by examining CDC reported case numbers in the US:

```
cdc <- data.frame(
                                    year = c(1922L,1923L,1924L,1925L,
                                              1926L,1927L,1928L,1929L,1930L,1931L,
                                              1932L,1933L,1934L,1935L,1936L,
                                              1937L,1938L,1939L,1940L,1941L,1942L,
                                              1943L,1944L,1945L,1946L,1947L,
                                              1948L,1949L,1950L,1951L,1952L,
                                              1953L,1954L,1955L,1956L,1957L,1958L,
                                              1959L,1960L,1961L,1962L,1963L,
                                              1964L,1965L,1966L,1967L,1968L,1969L,
                                              1970L,1971L,1972L,1973L,1974L,
                                              1975L,1976L,1977L,1978L,1979L,1980L,
                                              1981L,1982L,1983L,1984L,1985L,
                                              1986L,1987L,1988L,1989L,1990L,
                                              1991L,1992L,1993L,1994L,1995L,1996L,
                                              1997L,1998L,1999L,2000L,2001L,
                                              2002L,2003L,2004L,2005L,2006L,2007L,
                                              2008L,2009L,2010L,2011L,2012L,
                                              2013L,2014L,2015L,2016L,2017L,2018L,
                                              2019L,2020L,2021L,2022L,2023L, 2024L),
         cases = c(107473,164191,165418,152003,
                                              202210,181411,161799,197371,
                                              166914,172559,215343,179135,265269,
                                              180518,147237,214652,227319,103188,
                                              183866,222202,191383,191890,109873,
                                              133792,109860,156517,74715,69479,
                                              120718,68687,45030,37129,60886,
```

```
                                              62786,31732,28295,32148,40005,
                                              14809,11468,17749,17135,13005,6799,
                                              7717,9718,4810,3285,4249,3036,
                                              3287,1759,2402,1738,1010,2177,2063,
                                              1623,1730,1248,1895,2463,2276,
                                              3589,4195,2823,3450,4157,4570,
                                              2719,4083,6586,4617,5137,7796,6564,
                                              7405,7298,7867,7580,9771,11647,
                                              25827,25616,15632,10454,13278,
                                              16858,27550,18719,48277,28639,32971,
                                              20762,17972,18975,15609,18617,
                                              6124,2116,3044,7063, 22538)
       )
```

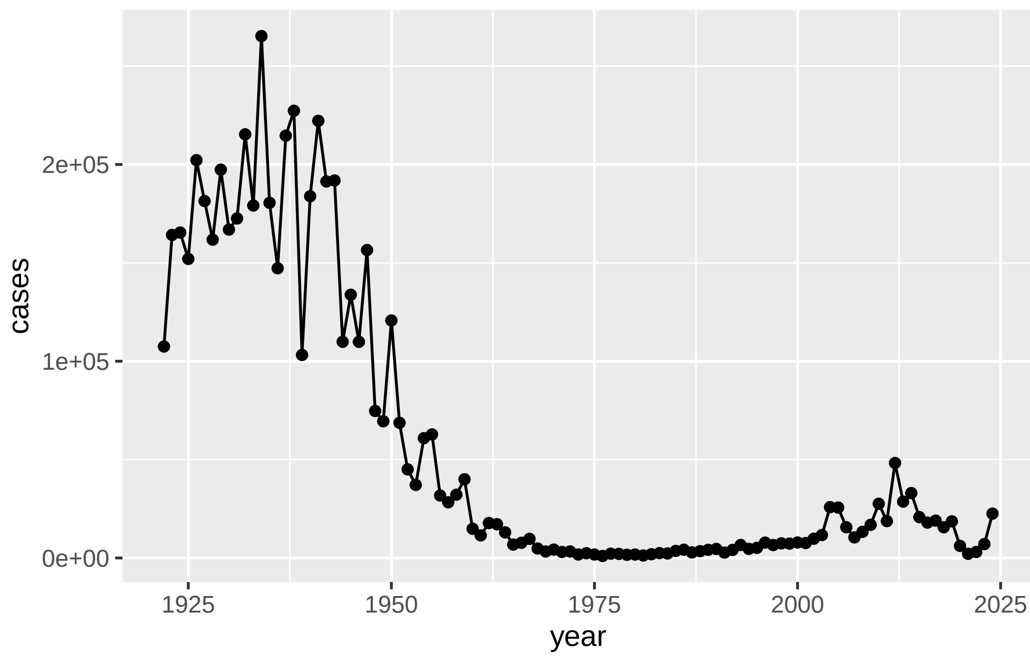
Plot of cases per year for Pertussis in the US

> Q1. With the help of the R "addin" package datapasta assign the CDC pertussis case number data to a data frame called cdc and use ggplot to make a plot of cases numbers over time.

```r
library(ggplot2)
```

```
Warning: package 'ggplot2' was built under R version 4.5.2
```

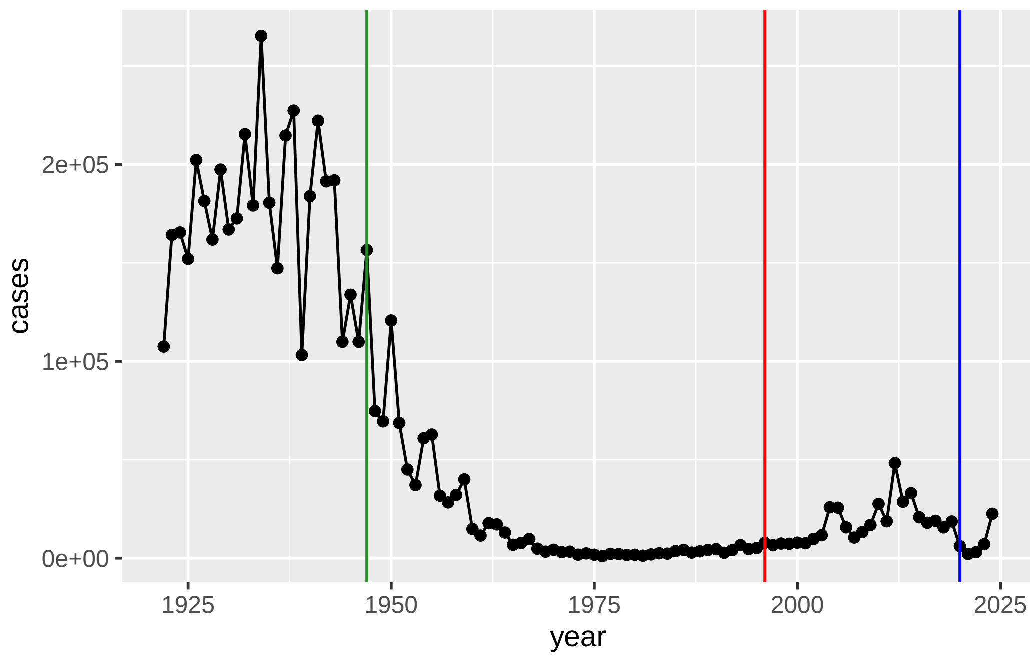
```r
ggplot(cdc) +
  aes(year, cases) +
  geom_point() +
  geom_line()
```



Add some major milestone timepoints to our plot:

> Q2. Using the ggplot geom_vline() function add lines to your previous plot for the 1946 introduction of the wP vaccine and the 1996 switch to aP vaccine (see example in the hint below). What do you notice?

```
ggplot(cdc) +
  aes(year, cases) +
  geom_point() +
  geom_line() +
  geom_vline(xintercept = 1947, col="forestgreen") +
  geom_vline(xintercept = 1996, col="red") +
  geom_vline(xintercept = 2020, col="blue")
```



> Q3. Describe what happened after the introduction of the aP vaccine? Do you have a possible explanation for the observed trend? The full introduction of the mandatory wP (whole-cell) Pertussis immunization in the mid 1940s led to a dramatic reduction in case numbers (from over 200,000 to 100s).

The switch to the aP (newer acellular formalization) had a lag period where people were switching from the wP to the aP vaccine that was followed by a large increase in case numbers with a peak of nearly 50,000 cases.

The 2020 lock-downs and social distancing measures showed a noticeable decrease in case numbers that slowly increased again once lockdown and social distancing was unregulated.

The CMI-PB project

The mission of CMI-PB is to provide the scientific community with a comprehensive, high-quality and freely accessible resource of Pertussis booster vaccination.

Website: hhttps://www.cmi-pb.org

They make their data available via JSON format API endpoints - basically the database tables in a key:value type format like "infancy_vac":"wP". To read this, we can use the `read_json()` function from the **jsonlite** package. Install with `install.packages("jsonlite)`

```r
library(jsonlite)
```

```
Warning: package 'jsonlite' was built under R version 4.5.2
```

```r
subject <- read_json(path="http://cmi-pb.org/api/v5_1/subject",
                     simplifyVector=TRUE)
head(subject)
```

```
  subject_id infancy_vac biological_sex               ethnicity  race
1          1          wP         Female Not Hispanic or Latino White
2          2          wP         Female Not Hispanic or Latino White
3          3          wP         Female                 Unknown White
4          4          wP           Male Not Hispanic or Latino Asian
5          5          wP           Male Not Hispanic or Latino Asian
6          6          wP         Female Not Hispanic or Latino White
  year_of_birth date_of_boost      dataset
1    1986-01-01    2016-09-12 2020_dataset
2    1968-01-01    2019-01-28 2020_dataset
3    1983-01-01    2016-10-10 2020_dataset
4    1988-01-01    2016-08-29 2020_dataset
5    1991-01-01    2016-08-29 2020_dataset
6    1988-01-01    2016-10-10 2020_dataset
```

> Q. How many "subjects"/individuals are in this dataset?

```r
nrow(subject)
```

```
[1] 172
```

> Q4. How many aP and wP subjects are there?

```r
table(subject$infancy_vac)
```

```

aP wP 
87 85 
```

> Q5. What is the breakdwon by "biolgical_sex" and "race"?

```r
table(subject$biological_sex)
```

```

Female   Male 
   112     60 
```

```r
table(subject$race)
```

```

            American Indian/Alaska Native 
                                        1 
                                    Asian 
                                       44 
                Black or African American 
                                        5 
                     More Than One Race 
                                       19 
Native Hawaiian or Other Pacific Islander 
                                        2 
                   Unknown or Not Reported 
                                       21 
                                    White 
                                       80 
```

Q6. What is the breakdown of race and biological sex (e.g. number of Asian females, White males etc…)?

```r
table(subject$race, subject$biological_sex)
```

```

                                            Female Male
  American Indian/Alaska Native                  0    1
  Asian                                         32   12
  Black or African American                      2    3
  More Than One Race                            15    4
  Native Hawaiian or Other Pacific Islander      1    1
  Unknown or Not Reported                       14    7
  White                                         48   32
```

This breakdown is not particularly representative of the US population - this is a serious cavet for this study. However, it is still the largest sample of its type ever assembled.

```r
specimen <- read_json("http://cmi-pb.org/api/v5_1/specimen", simplifyVector = TRUE)
ab_titer <- read_json("http://cmi-pb.org/api/v5_1/plasma_ab_titer", simplifyVector = TRUE)
```

```r
head(specimen)
```

```
  specimen_id subject_id actual_day_relative_to_boost
1           1          1                           -3
2           2          1                            1
3           3          1                            3
4           4          1                            7
5           5          1                           11
6           6          1                           32
  planned_day_relative_to_boost specimen_type visit
1                             0         Blood     1
2                             1         Blood     2
3                             3         Blood     3
4                             7         Blood     4
5                            14         Blood     5
6                            30         Blood     6
```

We need to "join" or link these tables with the `subject` table so we can begin to analyze this data and know who a given Ab sample was collected for and when.

> Q9. Complete the code to join specimen and subject tables to make a new merged data frame containing all specimen records along with their associated subject details:

```r
library(dplyr)
```

```

Attaching package: 'dplyr'


The following objects are masked from 'package:stats':

    filter, lag
```

```
The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
meta <- inner_join(subject, specimen)
```

```
Joining with `by = join_by(subject_id)`
```

```
head(meta)
```

```
  subject_id infancy_vac biological_sex              ethnicity  race
1          1          wP         Female Not Hispanic or Latino White
2          1          wP         Female Not Hispanic or Latino White
3          1          wP         Female Not Hispanic or Latino White
4          1          wP         Female Not Hispanic or Latino White
5          1          wP         Female Not Hispanic or Latino White
6          1          wP         Female Not Hispanic or Latino White
  year_of_birth date_of_boost      dataset specimen_id
1    1986-01-01    2016-09-12 2020_dataset           1
2    1986-01-01    2016-09-12 2020_dataset           2
3    1986-01-01    2016-09-12 2020_dataset           3
4    1986-01-01    2016-09-12 2020_dataset           4
5    1986-01-01    2016-09-12 2020_dataset           5
6    1986-01-01    2016-09-12 2020_dataset           6
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
2                            1                             1         Blood
3                            3                             3         Blood
4                            7                             7         Blood
5                           11                            14         Blood
6                           32                            30         Blood
  visit
1     1
2     2
3     3
4     4
5     5
6     6
```

Now let's join the `ab_titer` table with our `meta` table so we have all the information about a given Ab measurement

Q10. Now using the same procedure join meta with titer data so we can further analyze this data in terms of time of visit aP/wP, male/female etc.

```
ab_data <- inner_join(meta, ab_titer)
```

```
Joining with `by = join_by(specimen_id)`
```

```
head(ab_data)
```

```
  subject_id infancy_vac biological_sex              ethnicity  race
1          1          wP         Female Not Hispanic or Latino White
2          1          wP         Female Not Hispanic or Latino White
3          1          wP         Female Not Hispanic or Latino White
4          1          wP         Female Not Hispanic or Latino White
5          1          wP         Female Not Hispanic or Latino White
6          1          wP         Female Not Hispanic or Latino White
  year_of_birth date_of_boost      dataset specimen_id
1    1986-01-01    2016-09-12 2020_dataset           1
2    1986-01-01    2016-09-12 2020_dataset           1
3    1986-01-01    2016-09-12 2020_dataset           1
4    1986-01-01    2016-09-12 2020_dataset           1
5    1986-01-01    2016-09-12 2020_dataset           1
6    1986-01-01    2016-09-12 2020_dataset           1
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
2                           -3                             0         Blood
3                           -3                             0         Blood
4                           -3                             0         Blood
5                           -3                             0         Blood
6                           -3                             0         Blood
  visit isotype is_antigen_specific antigen        MFI MFI_normalised  unit
1     1     IgE               FALSE   Total 1110.21154       2.493425 UG/ML
2     1     IgE               FALSE   Total 2708.91616       2.493425 IU/ML
3     1     IgG                TRUE      PT   68.56614       3.736992 IU/ML
4     1     IgG                TRUE     PRN  332.12718       2.602350 IU/ML
5     1     IgG                TRUE     FHA 1887.12263      34.050956 IU/ML
6     1     IgE                TRUE     ACT    0.10000       1.000000 IU/ML
  lower_limit_of_detection
1                 2.096133
2                29.170000
3                 0.530000
```

```
4                  6.205949
5                  4.679535
6                  2.816431
```

Q. How many Ab (antibodies) measurements do we have in total?

```r
nrow(ab_data)
```

```
[1] 61956
```

Q. How many different isotypes (types of Ab) are in the dataset?

```r
unique(ab_data$isotype)
```

```
[1] "IgE"  "IgG"  "IgG1" "IgG2" "IgG3" "IgG4"
```

Q11. How many specimens (i.e. entries in abdata) do we have for each isotype?

```r
table(ab_data$isotype)
```

```
  IgE   IgG  IgG1  IgG2  IgG3  IgG4 
 6698  7265 11993 12000 12000 12000 
```

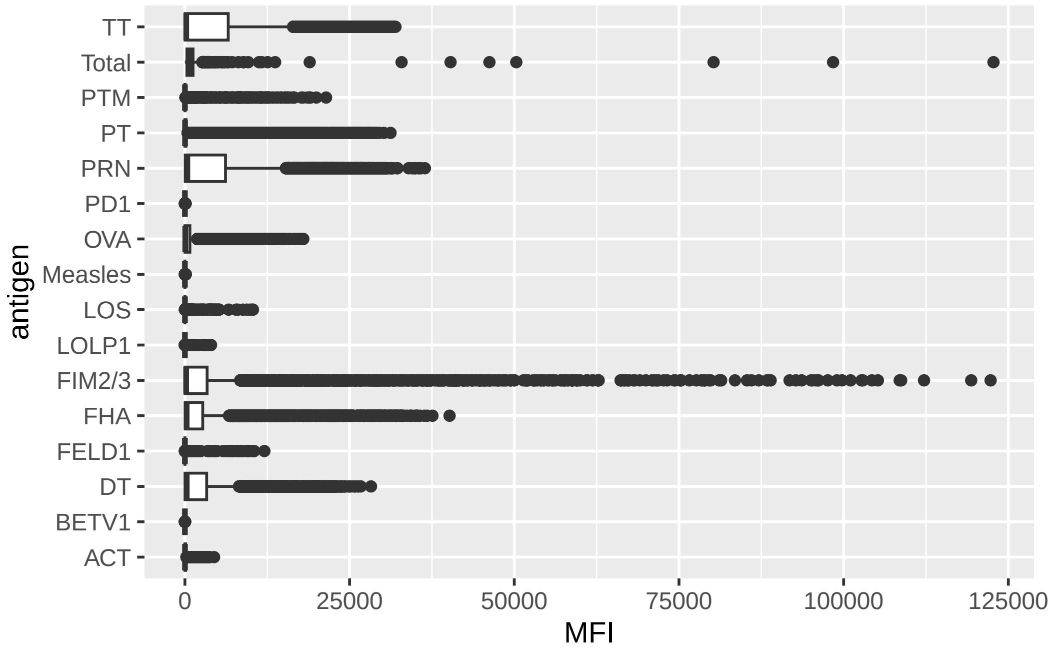
Q. How many different antigens?

```r
unique(ab_data$antigen)
```

```
 [1] "Total"   "PT"      "PRN"     "FHA"     "ACT"     "LOS"     "FELD1"  
 [8] "BETV1"   "LOLP1"   "Measles" "PTM"     "FIM2/3"  "TT"      "DT"     
[15] "OVA"     "PD1"    
```

```r
ggplot(ab_data) +
  aes(MFI, antigen) +
  geom_boxplot()
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_boxplot()`).
```



Q12. What are the different $dataset values in abdata and what do you notice about the number of rows for the most "recent" dataset?

```
table(ab_data$dataset)
```

```
2020_dataset 2021_dataset 2022_dataset 2023_dataset
       31520         8085         7301        15050
```

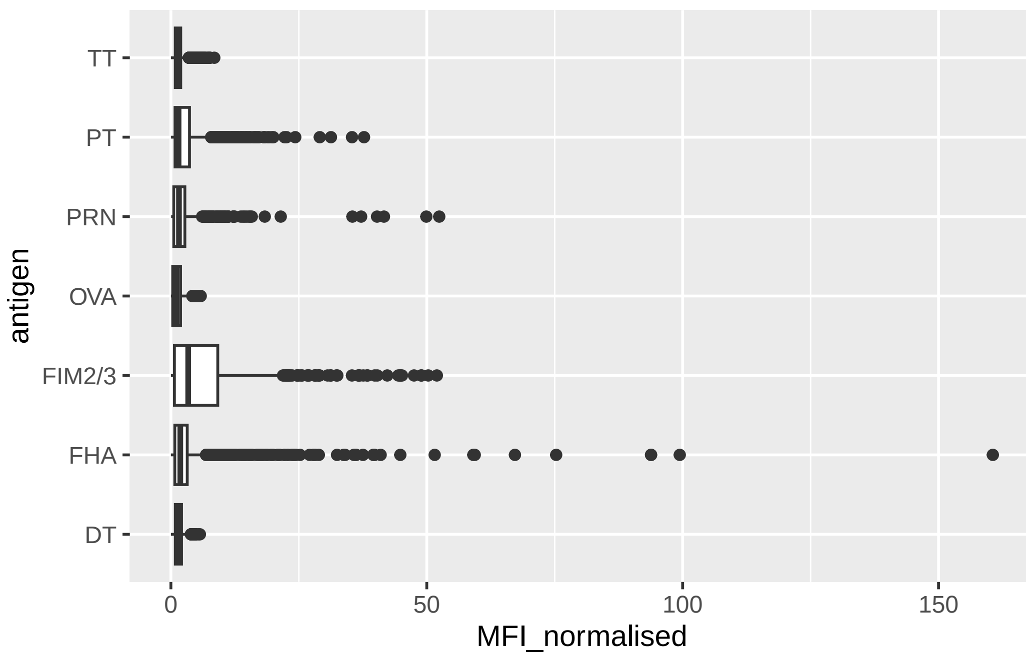
The most "recent" dataset has a much larger sample size compared to the previous years.

Examine IgG Ab titer levels

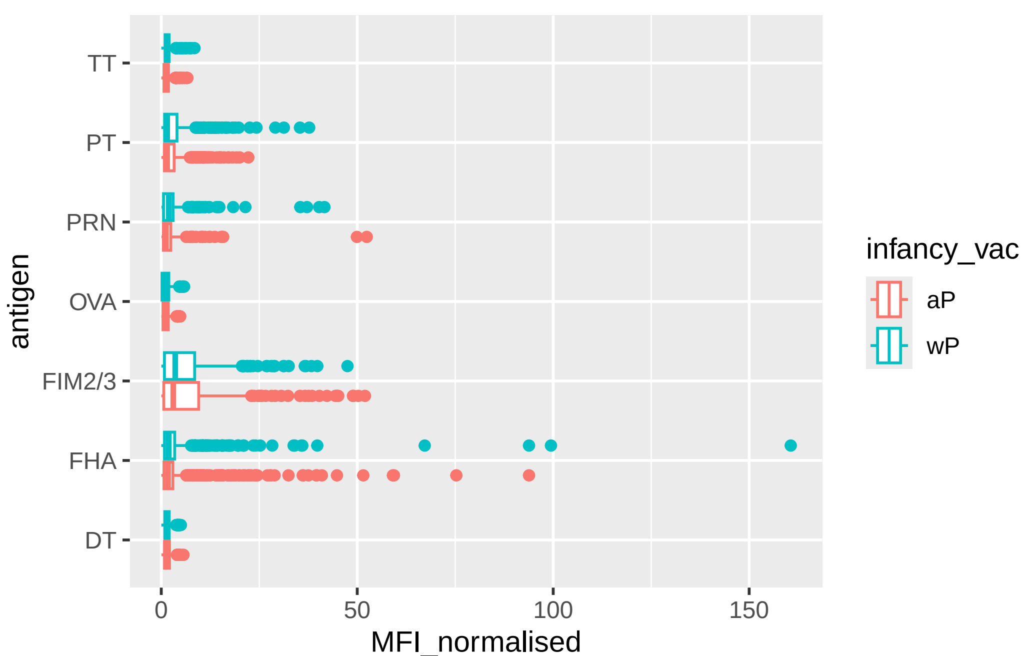
IgG is crucial for long-term immunity and responding to bacterial & viral infections.

```
igg <- ab_data |>
  filter(isotype=="IgG")
```

```
ggplot(igg) +
  aes(MFI_normalised, antigen) +
  geom_boxplot()
```

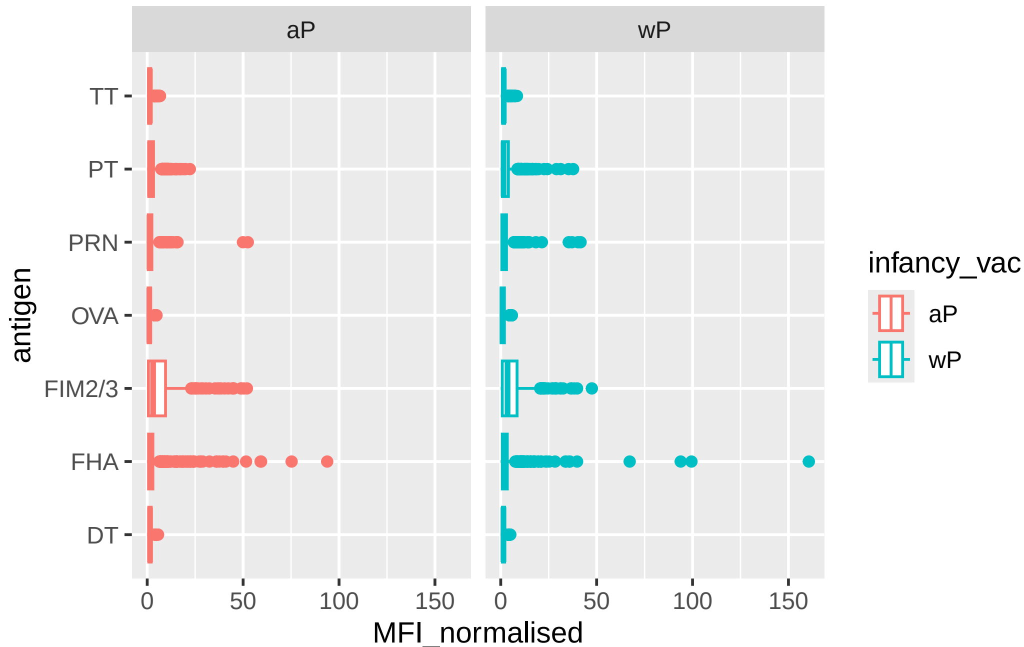



```r
ggplot(igg) +
  aes(MFI_normalised, antigen, col=infancy_vac) +
  geom_boxplot()
```



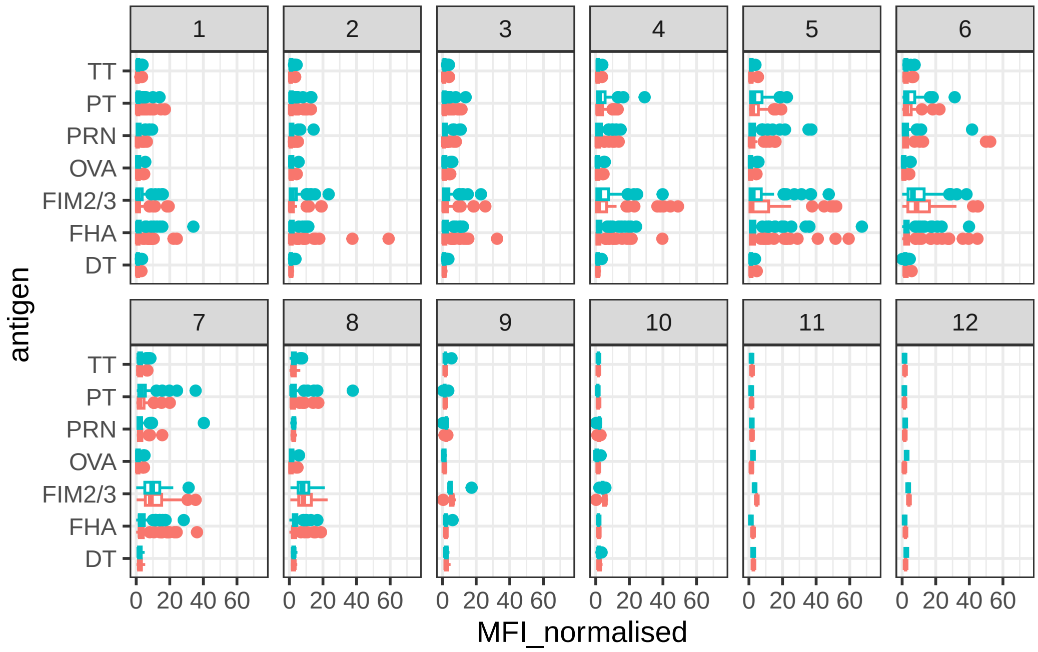
We can “facet” our plot by wP vs aP

```
ggplot(igg) +
  aes(MFI_normalised, antigen, col=infancy_vac) +
  geom_boxplot() +
  facet_wrap(~infancy_vac)
```



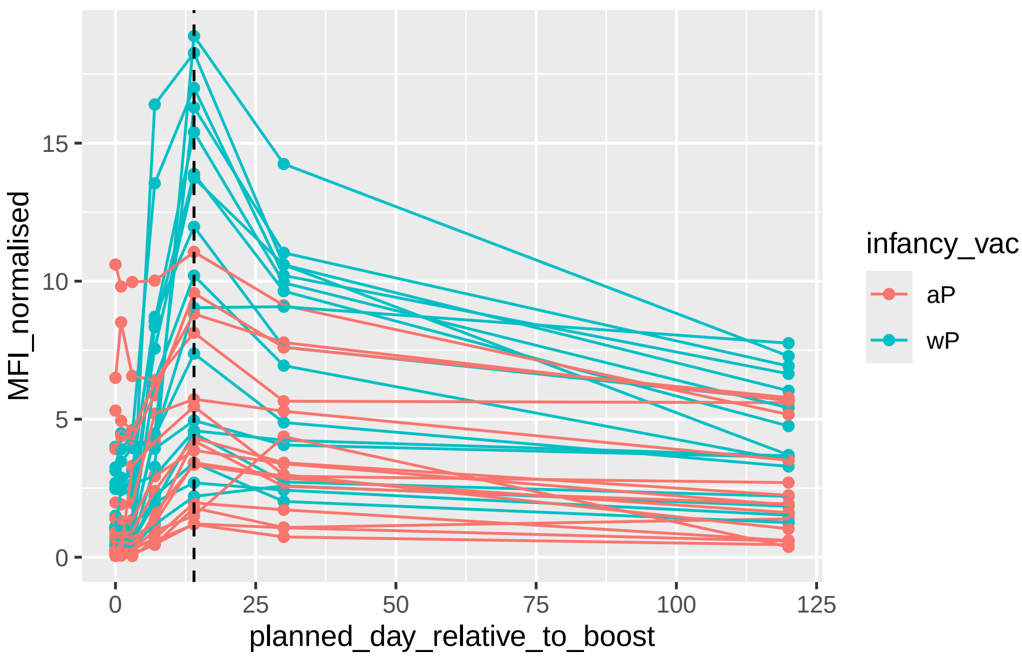

```
ggplot(igg) +
  aes(MFI_normalised, antigen, col=infancy_vac ) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(visit), nrow=2) +
  xlim(0,75) +
  theme_bw()
```

```
Warning: Removed 5 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```



More advanced analysis digging into individual antigen responses over time:

```
filter(igg, antigen == "PT", dataset == "2021_dataset") |>
 ggplot() +
 aes(x=planned_day_relative_to_boost,
       y=MFI_normalised,
       col=infancy_vac,
       group=subject_id) +
   geom_point() +
   geom_line() +
  geom_vline(xintercept=14, linetype="dashed")
```



This plot shows the time course of Pertussis toxin (PT) antibody responses for a large set of wP (teal color) and aP (red color) individuals. Levels peak at day 14 and are larger in magnitude for wP than aP individuals.

There are lots of cool things to explore in this dataset and we need coding and biology knowledge to do it effectively.